# Series 4: Applied Machine Learning and Predictive Modelling 1

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Load the `x-ray` dataset (file "Xray.RDS"). This study refers to a "study carried out to investigate the connection between X-ray usage and acute myeloid leukemia in childhood" (for further information see the help page about the `amlxray` dataset in `{faraway}` package). We work on a subset of the original data, in particular, we consider the following variables:

- `disease`: is the response variable (1 = *diseased*, 0 = *healthy*)
- `Sex`: Sex of the patient
- `age`: Age of the child (in years)
- `downs`: Presence of the Down syndrome
- `Mray`: Did the mother ever have an Xray
- `Fray`: Did the father ever have an Xray
- `CnRay.num`: Total number of Xrays of the child "1" = none; "2" = 1 or 2; "3" = 3 or 4; "4" = 5 or more

```r
d.xray <- readRDS("../../../Datasets/Xray.RDS")
str(d.xray)
```

```
Classes 'tbl_df', 'tbl' and 'data.frame':   112 obs. of  7 variables:
 $ Sex      : Factor w/ 2 levels "F","M": 1 2 1 2 2 1 1 2 1 2 ...
 $ disease  : num  1 1 0 1 1 1 1 0 0 0 ...
 $ age      : int  0 6 8 1 4 9 17 5 0 7 ...
 $ downs    : Factor w/ 2 levels "no","yes": 1 1 1 2 1 1 1 1 1 1 ...
 $ Mray     : Factor w/ 2 levels "no","yes": 1 1 1 1 2 2 1 1 1 1 ...
 $ Fray     : Factor w/ 2 levels "no","yes": 1 1 1 1 1 1 1 1 1 1 ...
 $ CnRay.num: num  1 3 1 1 1 1 2 1 1 1 ...
```

```r
head(d.xray)
```

```
  Sex disease age downs Mray Fray CnRay.num
1   F       1   0    no   no   no         1
2   M       1   6    no   no   no         3
3   F       0   8    no   no   no         1
4   M       1   1   yes   no   no         1
5   M       1   4    no  yes   no         1
6   F       1   9    no  yes   no         1
```

The response variable is *disease*, which is a binary variable. We will not use *downs* as predictor, but all the other variables can be included as predictors in your models.

The goal is to use 5-fold cross validation to compare four models. As a measure of predictive performance we want you to use the proportion of correctly classified observations for the test sample. The following questions will sequentially build the implementation of this goal.

## Question

*Fit the following four models on the whole dataset. Take a look at the summary-output of the four models. Compute the in-sample AIC-values (= AIC-values of the models fitted on the whole dataset) of all four models. You can use the function AIC(). Based on these in-sample AIC-values, which model would you select as the best performing model?*

```r
glm.xray.simple <- glm(formula = disease ~ age + Sex,
                       data = d.xray,
                       family = "binomial")
##
glm.xray.complex <- glm(formula = disease ~ age + Sex + Mray + Fray + CnRay.num,
                        data = d.xray,
                        family = "binomial")
##
glm.xray.very.complex <- glm(formula = disease ~ Sex * (poly(age, degree = 2) +
                                                        Mray + Fray + CnRay.num) ,
                        data = d.xray,
                        family = "binomial")
## we need to load the package mgcv to use the function gam()
library(mgcv)
gam.xray <- gam(formula = disease ~ s(age, by = Sex) + Sex + Mray + Fray + CnRay.num,
                data = d.xray,
                family = "binomial")
```

## Question

*Calculate the proportion of correctly classified observations in-sample (that is on the whole data set as well) for all four models. Use a cutoff of 0.5 in the predicted probability for disease. This means that if a patient has a predicted probability of 0.5 or higher, we would predict disease = 1 for this patient. Based on the in-sample proportion of correctly classified observations, which model would you choose?*

Hint: Use type = "response" when predicting from a glm/gam with binary response, to obtain the predicted probabilities. You can write a function that has as two arguments the "model" and the "newdata" to compute the proportion of correctly classified observations. We can then reuse this function later.

## Question

*Now implement 5-fold CV to compare the estimated proportion of correctly classified observations on test data. If you want, you can implement repeated 5-fold CV to see how much the estimated proportion of correctly classified observations varies between different 5-fold CVs (different permutation of the rows of the data).*

## Question

*Based on the results of the 5-fold CV, which of the four models would you choose and why?*

## Question

*Ask generative AI about the evaluation of the predictive performance of models and critically compare the obtained answer with what you learned in this class.*